Huiju (Heejoo) Lee

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LinkedIn | Google Scholar | ResearchGate | GitHub

Ph.D. candidate specializing in computational materials science, atomistic simulations, and AI for materials discovery. Experienced in developing machine-learning interatomic potentials, performing large-scale materials simulations, and building high-throughput computational workflows. Strong background in density functional theory, molecular dynamics, Python, and high-performance computing.

Education

Ph.D. Mechanical Engineering, Portland State University, OR, USA | Tau Beta Pi	GPA 4.0/4.0	*Expected Mar 2027*
M.S. Materials Processing & Engineering, Inha University, Incheon, South Korea	GPA 4.0/4.0	*Feb 2023*
M.S. Mechanical Engineering, Portland State University, OR, USA | Phi Kappa Phi	GPA 4.0/4.0	*Mar 2022*
B.S. Materials Science & Engineering, Inha University, Incheon, South Korea	GPA 3.67/4.0	*Feb 2020*

Skills & Tools

Atomistic Simulations: DFT (VASP, QuantumATK, GPAW, Quantum Espresso, SIESTA); MD(LAMMPS); MLIPs; Phonopy
Materials Informatics: Materials Project, OQMD, ICSD; ASE, pymatgen; materials screening / property prediction
Programming & HPC: Python, Bash, Linux; SLURM, Flux; MPI, OpenMP; high-throughput computing
Machine Learning & AI: Deep Learning, Graph Neural Networks (GNNs); PyTorch; statistical analysis, hyperparameter tuning
Semiconductor & Materials: Hands-on cleanroom training in photolithography, thin-film deposition; SEM/EDS, EBSD, XRD, AFM

Experience

- **Lawrence Livermore National Laboratory, Livermore, CA (Remote)** – Research Intern | *June 2026 – August 2026*
  - Developed atomistic modeling workflows integrating density functional theory (DFT), molecular dynamics (MD), and machine-learning interatomic potentials (MLIPs) for molten-salt nuclear reactor materials
  - Conducted convergence tests and parameter tuning, performed **600+ DFT calculations on HPC** systems to generate training data, and trained and evaluated MLIP models for electrochemical interface simulations

- **Portland State University, Portland, OR** – Graduate Research Assistant | *January 2023 – Present*
  - Developed **high-throughput DFT and machine-learning workflows** for large-scale simulation and data generation.
  - Generated and released a **large-scale DFT/phonon dataset** (2,738 materials; 77 elements; 15,670 supercells) and trained a **universal MACE MLIP**, achieving 0.18 THz phonon MAE and 86.2% dynamical-stability accuracy.
  - Developed high-throughput workflows for **temperature-dependent phonons and lattice thermal conductivity**, including MLIP fine-tuning and large-scale materials screening.
  - Collaborated with **experimental research teams at Yale and Penn State University**, providing computational insights into structural stability and thermal conductivity of two-dimensional chalcogenides and thermoelectric materials.

- **Inha University, South Korea** – Graduate Research Assistant | *March 2021 – January 2023*
  - Led **microstructural and thermomechanical analyses** of 1,600 MPa-class automotive steels in a 9-institution consortium, collaborating with multiple steel companies; evaluated and **optimized heat-treatment conditions using SEM/EDS and EBSD**.

- **Portland State University, Portland, OR** – Graduate Research Assistant | *March 2020 – March 2021*
  - Performed electron-transport simulations of **nanoscale semiconductor interconnect materials** using QuantumATK, with guidance from **Dr. Sanjay Natarajan, then Senior Vice President at Intel**.
  - Identified a Cu-functionalized carbon nanotube (CNT) structure with **~5× higher current at 1.0 V** than pristine CNT.

- **Inha University, South Korea** – Undergraduate Researcher | *Jul 2019 – Dec 2019*
  - Trained in materials characterization using SEM/EDS, EBSD, TEM, XRD, XPS, AFM, and Raman spectroscopy.
  - Completed hands-on cleanroom training in semiconductor fabrication.

Publications & Conference Presentations

- 15 peer-reviewed publications | 208+ citations | h-index 7 | i10-index 5 | 10 conference presentations

- Selected publication (Full list: Google Scholar):
  - **Huiju Lee***, Zhi Li*, Jiangang He, Yi Xia. (2026). Data-Driven Exploration and Insights into Temperature-Dependent Phonons in Inorganic Materials. Small, e00071.
  - **Huiju Lee**, Vinay I. Hegde, Chris Wolverton, Yi Xia. (2025). Accelerating high-throughput phonon calculations via machine learning universal potentials. Materials Today Physics, 53, 101688.
  - **Huiju Lee**, Yi Xia. (2024). Machine learning a universal harmonic interatomic potential for predicting phonons in crystalline solids. Applied Physics Letters, 124(10), 102202.
  - Zhi Li*, **Huiju Lee***, Chris Wolverton, Yi Xia. (2025). High-throughput computational framework for high-order anharmonic thermal transport in cubic and tetragonal crystals. npj Computational Materials, 12(1), 51. (*Co-first authors)

Academic Service

- Ad Hoc Reviewer: npj Computational Materials, Scientific Reports
- Undergraduate Research Mentor (2023-2025): Mentored undergraduate researchers in ML model training, evaluation, and hyperparameter optimization for materials science applications.

Fellowship & Awards

- Outstanding Research Assistant Award, Portland State University, Jun 2026
- **$60K+ Overseas Research Fellowship**, KIAT (Korea Institute for Advancement of Technology), Mar 2020 – Mar 2021
- National Scholarship for Science and Engineering, Korea Student Aid Foundation, 2018-2019
- Research Presentation Award, Korea Iron & Steel Association, in collaboration with POSCO, Oct 2019
- Research Presentation Award, Inha University, Nov 2017

Selected Certifications

- Surface Chemistry: Molecular Modeling for Materials Science Applications – Schrödinger, Jan 2026
- **Advanced Topics in AI for Science** – Argonne National Laboratory, Dec 2025
- **AI for Science on Supercomputers** – Argonne National Laboratory, Dec 2024
- Semiconductor Materials and Manufacturing – Portland State University, Mar 2022
- Advanced Data Analytics Certification – Korea Data Agency, Dec 2021